Stochastic simulation of chemical and non-chemical systems using Gillespie's algorithm

Aryan Gupta
4th year BS-MS
Department of Biological sciences

Indian Institute of Science Education and Research Bhopal

open_board

Contents

1. Introduction
2. Master Equations
3. Stochastic Simulation Algorithm (SSA)
4. Simulation Results
5. Summary

open_board

Table of Contents

1. Introduction
2. Master Equations
3. Stochastic Simulation Algorithm (SSA)
4. Simulation Results
5. Summary

The Problem

Assumptions :

- Fixed volume.
- Spatially homogeneous (Well-stirred).
- N chemical species and their respective counts at $t = 0$.
- M reactions

Goal :

To find the molecular population of all N species at any time $t > 0$

The Problem

Assumptions :

- Fixed volume.
- Spatially homogeneous (Well-stirred).
- N chemical species and their respective counts at $t = 0$.
- M reactions

Goal :

To find the molecular population of all N species at any time $t > 0$

What is Stochasticity?

- Deterministic : Completely determined by initial conditions and the "rules" of the system, No randomness, exact identical outcomes for the same input always.
- Stochastic: Not completely predictable given initial conditions, some element of randomness, probabilistically certain, input in a neighbourhood $\Rightarrow$ output in some neighbourhood.
- Chaotic: Extremely sensitive to initial conditions, even though not undeterministic. Input in a neighbourhood $\not\Rightarrow$ output in some neighbourhood

open_board

What is Stochasticity?

- Deterministic : Completely determined by initial conditions and the "rules" of the system, No randomness, exact identical outcomes for the same input always.
- Stochastic: Not completely predictable given initial conditions, some element of randomness, probabilistically certain, input in a neighbourhood $\Rightarrow$ output in some neighbourhood.
- Chaotic: Extremely sensitive to initial conditions, even though not undeterministic. Input in a neighbourhood $\not\Rightarrow$ output in some neighbourhood

open_board

What is Stochasticity?

- Deterministic : Completely determined by initial conditions and the "rules" of the system, No randomness, exact identical outcomes for the same input always.
- Stochastic: Not completely predictable given initial conditions, some element of randomness, probabilistically certain, input in a neighbourhood $\Rightarrow$ output in some neighbourhood.
- Chaotic: Extremely sensitive to initial conditions, even though not undeterministic. Input in a neighbourhood $\not\Rightarrow$ output in some neighbourhood

open_board

Deterministic approach

- Assumes reactions are continuous and deterministic, uses ODEs.
- $\mathbf{X} = (X_1, X_2, \cdots X_n)$ represents the state vector of the system for n species.
- Time evolution of $\mathbf{X}$ is given by sysem of equations :
  $\dot{X}_i = f_i(X_1, X_2, \cdots X_n)$
- Deterministic only in very high dimensional phase spaces like $\approx 10^{23}$.
- Conceals(*statistically, averages out*) many properties of systems of the size $\approx 10^2 - 10^5$

open_board

Deterministic approach

- Assumes reactions are continuous and deterministic, uses ODEs.
- $\mathbf{X} = (X_1, X_2, \cdots X_n)$ represents the state vector of the system for n species.
- Time evolution of $\mathbf{X}$ is given by sysem of equations :
  $\dot{X}_i = f_i(X_1, X_2, \cdots X_n)$
- Deterministic only in very high dimensional phase spaces like $\approx 10^{23}$.
- Conceals(*statistically, averages out*) many properties of systems of the size $\approx 10^2 - 10^5$

open_board

Deterministic approach

- Assumes reactions are continuous and deterministic, uses ODEs.
- $\mathbf{X} = (X_1, X_2, \cdots X_n)$ represents the state vector of the system for n species.
- Time evolution of $\mathbf{X}$ is given by sysem of equations :
  $\dot{X}_i = f_i(X_1, X_2, \cdots X_n)$
- Deterministic only in very high dimensional phase spaces like $\approx 10^{23}$.
- Conceals(*statistically, averages out*) many properties of systems of the size $\approx 10^2 - 10^5$

open_board

Deterministic approach

- Assumes reactions are continuous and deterministic, uses ODEs.
- $\mathbf{X} = (X_1, X_2, \cdots X_n)$ represents the state vector of the system for n species.
- Time evolution of $\mathbf{X}$ is given by sysem of equations :
  $\dot{X}_i = f_i(X_1, X_2, \cdots X_n)$
- Deterministic only in very high dimensional phase spaces like $\approx 10^{23}$.
- Conceals(*statistically, averages out*) many properties of systems of the size $\approx 10^2 - 10^5$

open_board

Deterministic approach

- Assumes reactions are continuous and deterministic, uses ODEs.
- $\mathbf{X} = (X_1, X_2, \cdots X_n)$ represents the state vector of the system for n species.
- Time evolution of $\mathbf{X}$ is given by sysem of equations :
  $\dot{X}_i = f_i(X_1, X_2, \cdots X_n)$
- Deterministic only in very high dimensional phase spaces like $\approx 10^{23}$.
- Conceals(*statistically, averages out*) many properties of systems of the size $\approx 10^2 - 10^5$

open_board

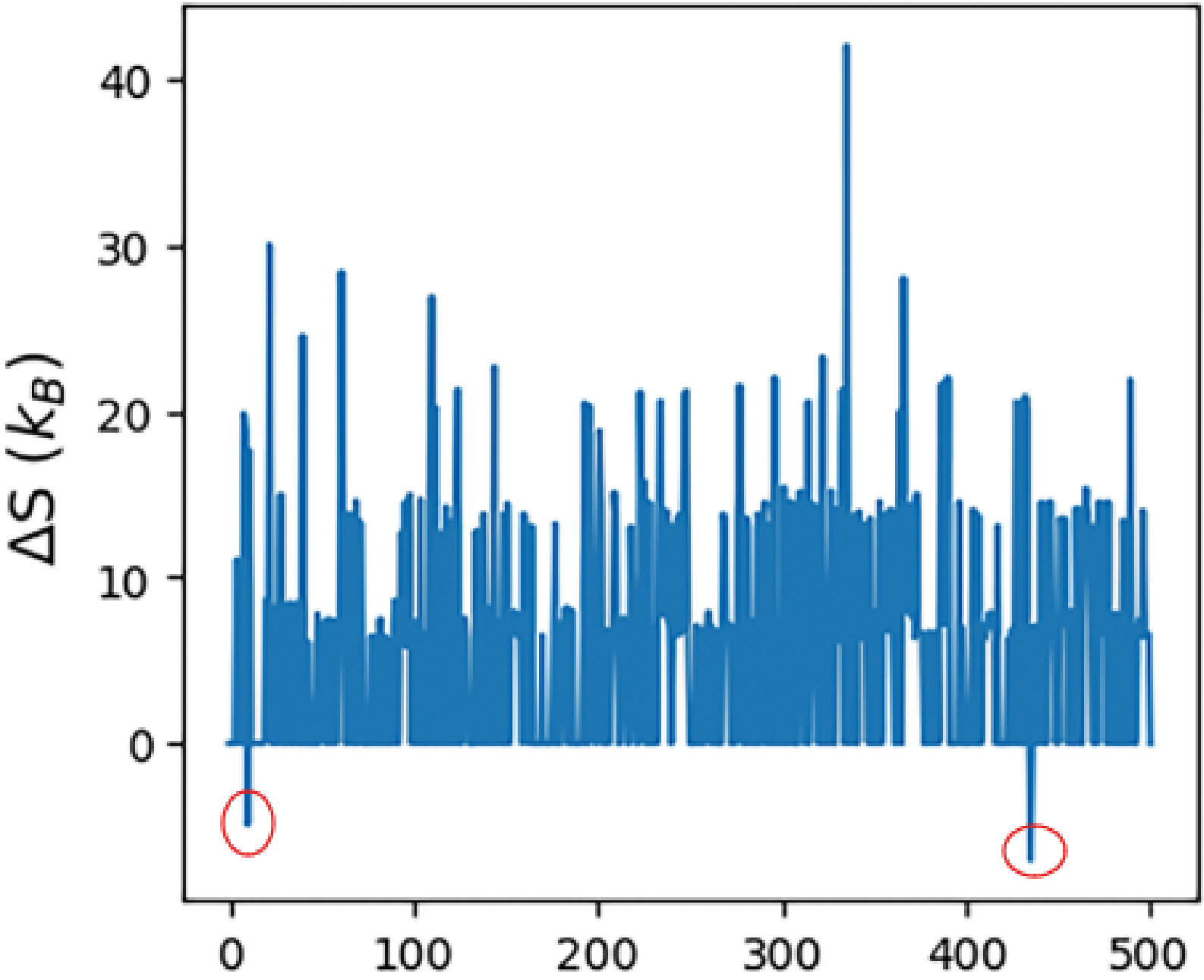

40
30
20
10
0
$\Delta S\ (k_B)$
0
100
200
300
400
500

Table of Contents

1. Introduction
2. Master Equations
3. Stochastic Simulation Algorithm (SSA)
4. Simulation Results
5. Summary

Master Equations

- Treats the system as stochastic, type of difference-differnetial equation.
- Describe the time evolution of a system's probability distribution rahter than time evolution of $\mathbf{X} = (X_1, X_2, \cdots X_n)$
- We will be concernedd with only two types of master equatios: CME nad RDME.

Master Equations

- Treats the system as stochastic, type of difference-differnetial equation.
- Describe the time evolution of a system's probability distribution rahter than time evolution of $\mathbf{X} = (X_1, X_2, \cdots X_n)$
- We will be concernedd with only two types of master equatios: CME nad RDME.

Master Equations

- Treats the system as stochastic, type of difference-differnetial equation.
- Describe the time evolution of a system's probability distribution rahter than time evolution of $\mathbf{X} = (X_1, X_2, \cdots X_n)$
- We will be concernedd with only two types of master equatios: CME nad RDME.

open_board

CME : Chemical master equation

- $\frac{\mathrm{d}P(\mathbf{x},t)}{\mathrm{d}t} = \mathfrak{R}P(\mathbf{x},t) =$
- $\sum_{\nu}^{V}\sum_{r}^{R}\left[-a_r(\mathbf{x}_\nu)P(\mathbf{x}_\nu,t) + a_r(\mathbf{x}_\nu - \mathbf{S}_r)P(\mathbf{x}_\nu - \mathbf{S}_r,t)\right]$
- ν is the index for a single state of the system
- r is the reaction index
- $a_r(\mathbf{x}_\nu)$ is the probability per unit time for reaction r to happen given a particular state $\mathbf{x}_\nu$
- $\mathbf{S}_r$ is the stoichiometric vector for the reaction r

open_board

RDME : Reaction Diffusion master equation

- $\frac{\mathrm{d}P(\mathbf{x},t)}{\mathrm{d}t} = \mathfrak{R}P(\mathbf{x},t) + \mathfrak{D}P(\mathbf{x},t)$
- $\mathfrak{D}P(\mathbf{x},t) =$

$$\sum_{\nu}^{V} \sum_{\xi=\pm\hat{i},\hat{j},\hat{k}} \sum_{\alpha}^{N} -d^{\alpha} x_{\nu}^{\alpha} P(\mathbf{x},t) + d^{\alpha}(x_{\nu+\xi}^{\alpha}+1)P(\mathbf{x}+1_{\xi+\alpha}^{\alpha}-1_{\nu}^{\alpha},t)$$

- ξ is $\pm\hat{i},\hat{j},\hat{k}$
- d^{α} is the diffusion constant for species α
- x_{ν}^{α} is the number of species α in voxel ν

open_board

Table of Contents

1. Introduction
2. Master Equations
3. Stochastic Simulation Algorithm (SSA)
4. Simulation Results
5. Summary

The Gillespie Algorithm

- A Monte Carlo method for stochastic simulation, generates statistically correct trajectories.
- Uses "propensity functions" to govern reaction rates
- two random numbers generated uniformally from $(0, 1]$
- r_1 determines when the next reaction/collission will happen
- r_2 determines what reaction will happen
- Demonstration will help...

open_board

The Gillespie Algorithm

- A Monte Carlo method for stochastic simulation, generates statistically correct trajectories.
- Uses "propensity functions" to govern reaction rates
- two random numbers generated uniformally from $(0, 1]$
- r_1 determines when the next reaction/collission will happen
- r_2 determines what reaction will happen
- Demonstration will help...

open_board

The Gillespie Algorithm

- A Monte Carlo method for stochastic simulation, generates statistically correct trajectories.
- Uses "propensity functions" to govern reaction rates
- two random numbers generated uniformally from $(0, 1]$
- r_1 determines when the next reaction/collission will happen
- r_2 determines what reaction will happen
- Demonstration will help...

open_board

The Gillespie Algorithm

- A Monte Carlo method for stochastic simulation, generates statistically correct trajectories.
- Uses "propensity functions" to govern reaction rates
- two random numbers generated uniformally from $(0, 1]$
- r_1 determines when the next reaction/collission will happen
- r_2 determines what reaction will happen
- Demonstration will help...

The Gillespie Algorithm

- A Monte Carlo method for stochastic simulation, generates statistically correct trajectories.
- Uses "propensity functions" to govern reaction rates
- two random numbers generated uniformally from $(0, 1]$
- r_1 determines when the next reaction/collission will happen
- r_2 determines what reaction will happen
- Demonstration will help...

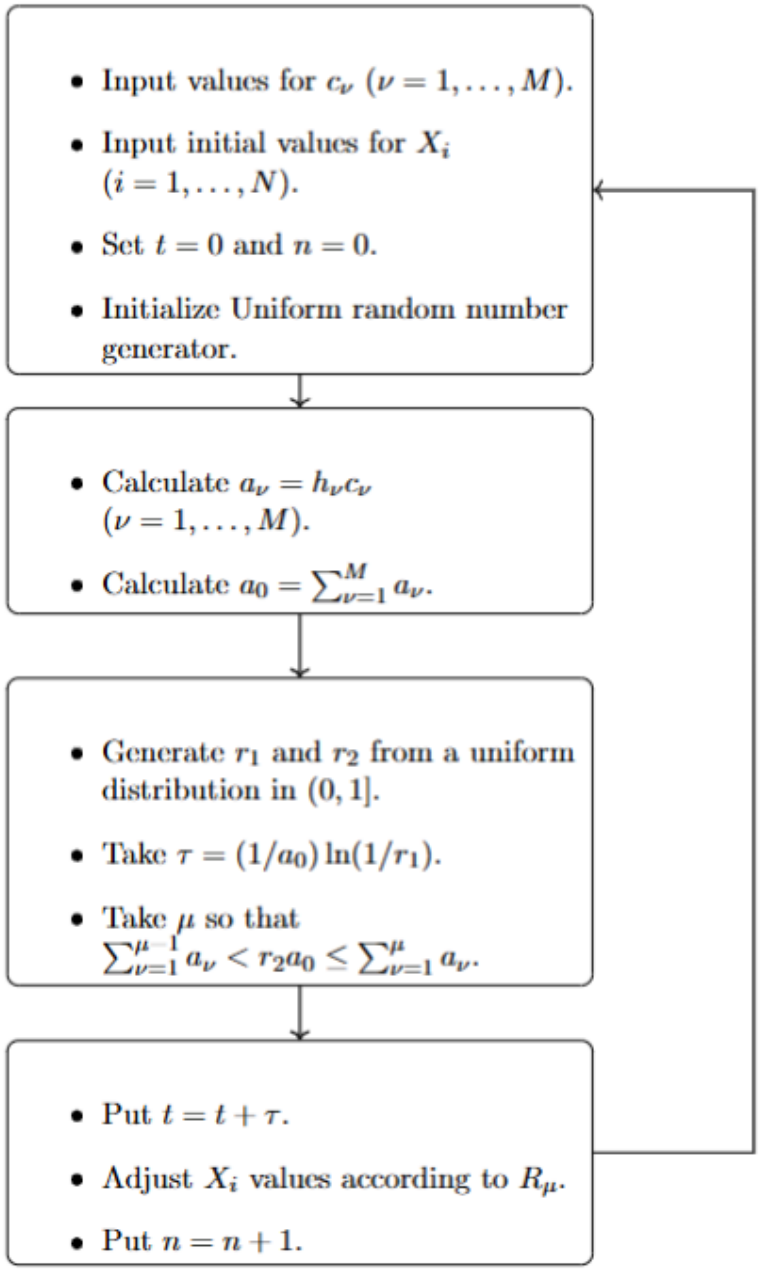
open_board

The Gillespie Algorithm

- A Monte Carlo method for stochastic simulation, generates statistically correct trajectories.
- Uses "propensity functions" to govern reaction rates
- two random numbers generated uniformally from $(0, 1]$
- r_1 determines when the next reaction/collission will happen
- r_2 determines what reaction will happen
- Demonstration will help...

open_board

SSA Flowchart



open_board

Fixed point analysis



SSA Advantages

- **Exactness**: No fixed time steps as in the case of numerical simulation of ODEs
- **Stochasticity**: Inherently models randomness of the collissions.
- **Generality**: Works for general systems, chemical and non-chemical.

open_board

Table of Contents

1. Introduction
2. Master Equations
3. Stochastic Simulation Algorithm (SSA)
4. Simulation Results
5. Summary

Bimolecular Reactions

- A simple reaction: $A + B \rightleftharpoons C$.
- Fluctuations are visible in the stochastic simulation.
- Averaging multiple stochastic runs approaches the deterministic result.

open_board

Bimolecular Reactions

- A simple reaction: $A + B \rightleftharpoons C$.
- Fluctuations are visible in the stochastic simulation.
- Averaging multiple stochastic runs approaches the deterministic result.

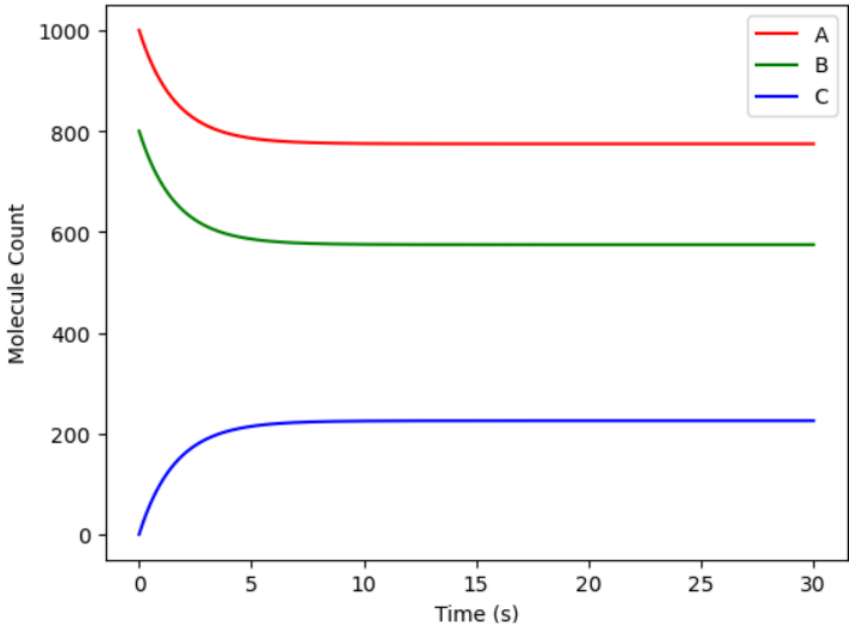
open_board

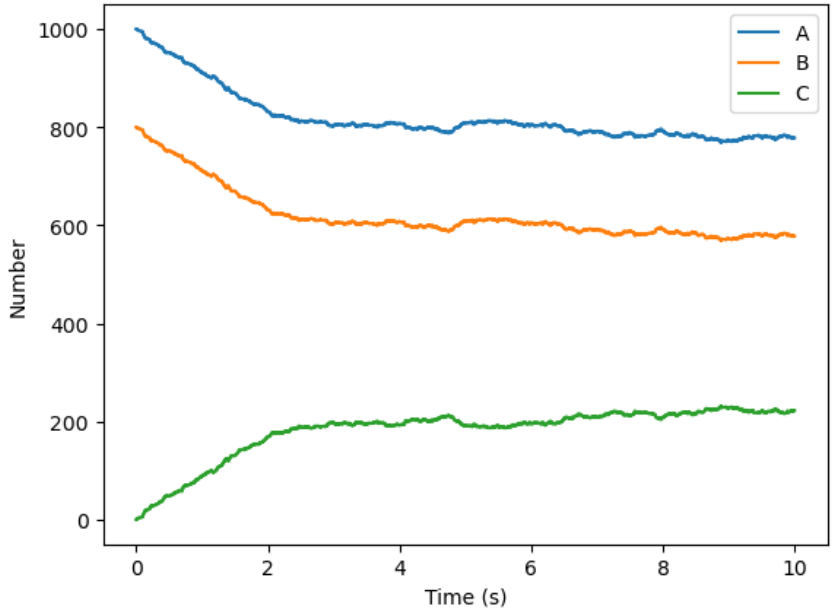
Bimolecular Reactions

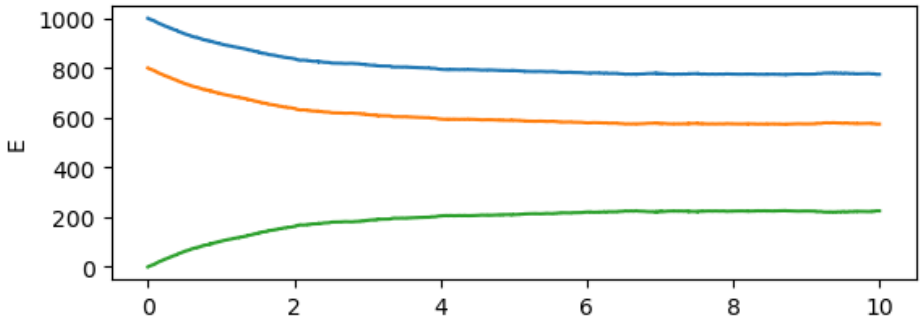
- A simple reaction: $A + B \rightleftharpoons C$.
- Fluctuations are visible in the stochastic simulation.
- Averaging multiple stochastic runs approaches the deterministic result.

open_board

Comparison of Simulation Results







open_board

Michaelis-Menten Kinetics

- $E + S \underset{k_{off}}{\overset{k_{on}}{\rightarrow}} ES \xrightarrow{k_{cat}} P$
- pyLM simulation data closely fits the theoretical curve.

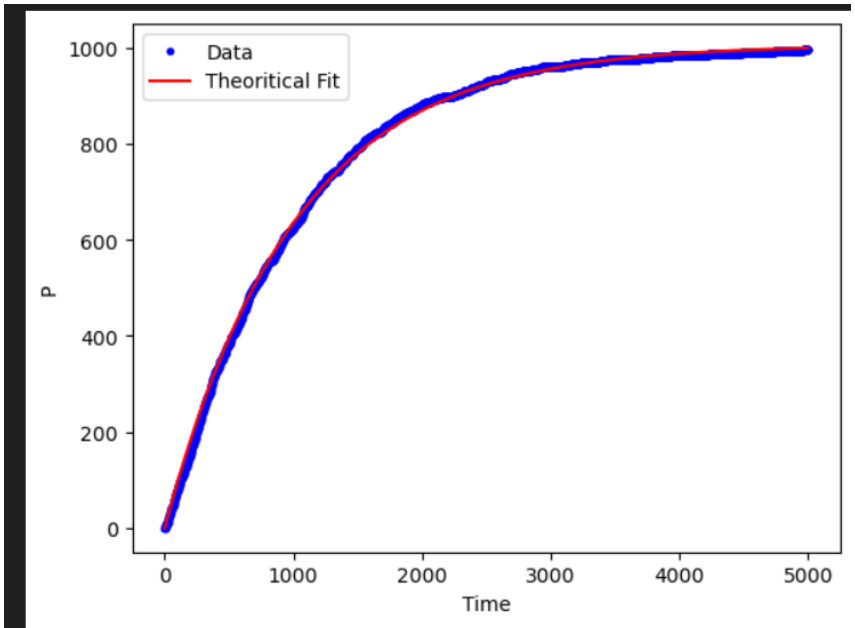
Michaelis-Menten Kinetics

- $E + S \underset{k_{off}}{\overset{k_{on}}{\rightleftarrows}} ES \xrightarrow{k_{cat}} P$
- pyLM simulation data closely fits the theoretical curve.

open_board

Michaelis-Menten Kinetics

- $E + S \underset{k_{off}}{\overset{k_{on}}{\rightarrow}} ES \xrightarrow{k_{cat}} P$
- pyLM simulation data closely fits the theoretical curve.



Lotka-Volterra Systems

$$\frac{dx}{dt} = \alpha x - \beta xy$$
$$\frac{dy}{dt} = -\gamma y + \delta xy$$

- Predator-prey ecological model.
- Stochastic simulations show characteristic population oscillations.
- Behavior matches the expected phase space diagram.

open_board

Lotka-Volterra Systems

$$\frac{dx}{dt} = \alpha x - \beta xy$$
$$\frac{dy}{dt} = -\gamma y + \delta xy$$

- Predator-prey ecological model.
- Stochastic simulations show characteristic population oscillations.
- Behavior matches the expected phase space diagram.

open_board

Lotka-Volterra Systems

$$\frac{dx}{dt} = \alpha x - \beta xy$$
$$\frac{dy}{dt} = -\gamma y + \delta xy$$

- Predator-prey ecological model.
- Stochastic simulations show characteristic population oscillations.
- Behavior matches the expected phase space diagram.

open_board

Stability Analysis

1. Identify fixed points by setting $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$.
2. Linearize the system around fixed points.
3. Analyze eigenvalues of the Jacobian matrix.

open_board

Stability Analysis

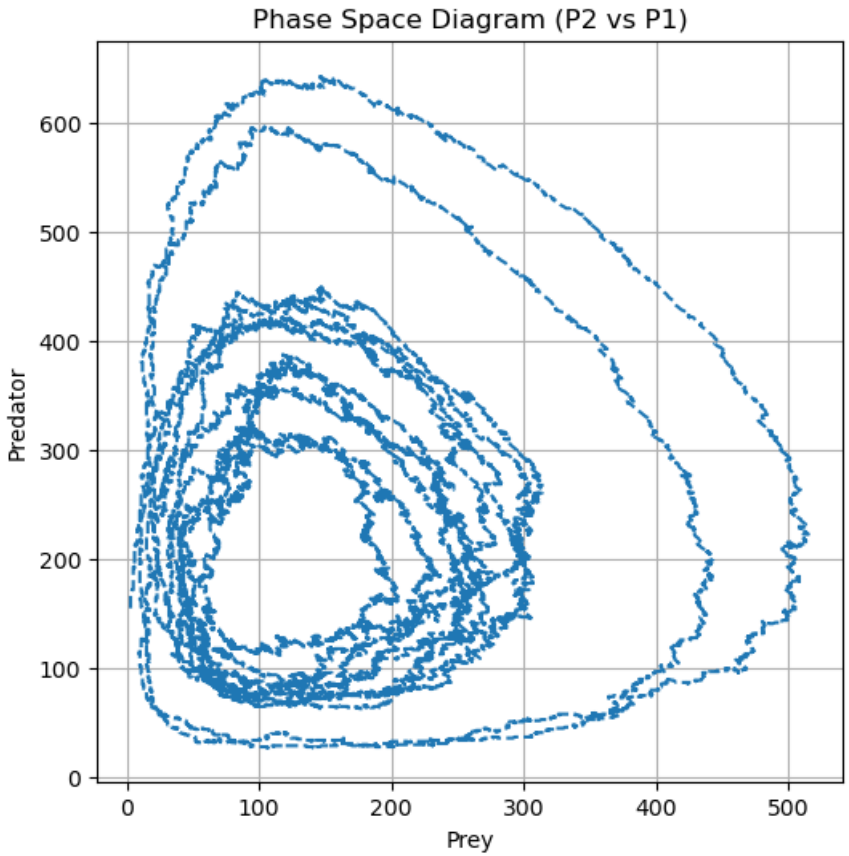
1. Identify fixed points by setting $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$.
2. Linearize the system around fixed points.
3. Analyze eigenvalues of the Jacobian matrix.

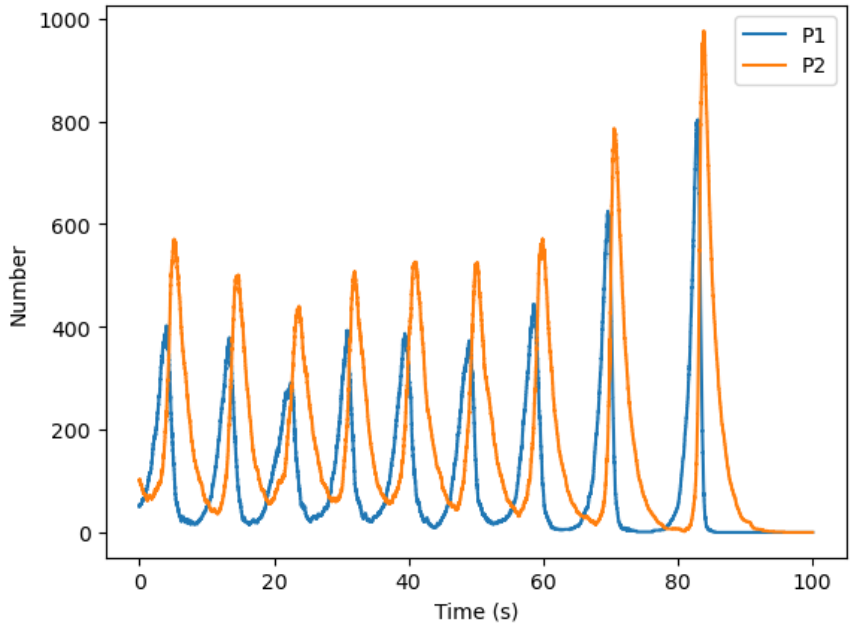
open_board

Stability Analysis

1. Identify fixed points by setting $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$.
2. Linearize the system around fixed points.
3. Analyze eigenvalues of the Jacobian matrix.

open_board


Phase Space Diagram (P2 vs P1)
Predator
Prey
0
100
200
300
400
500
600


P1
P2
Number
Time (s)
0
200
400
600
800
1000
0
20
40
60
80
100

open_board

min System

- `minDadp` – MinD protein with an ADP molecule bound
- `minDatp` – MinD protein with an ATP molecule bound
- `minDm` – Membrane-bound MinD protein
- `minE` – MinE protein
- `minDEm` – Membrane-bound MinD protein also bound to a MinE protein

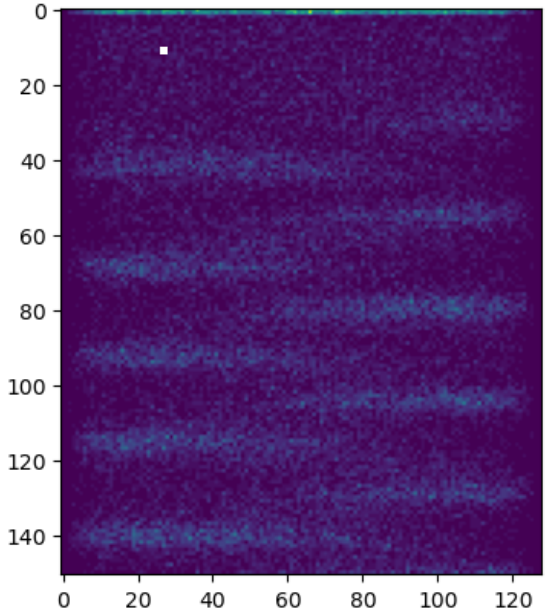


Table of Contents

1. Introduction
2. Master Equations
3. Stochastic Simulation Algorithm (SSA)
4. Simulation Results
5. Summary

Summary

- Stochastic methods are essential for biological systems(especially small systems wehre deterministic models don't work reliably)
- pyLM is a powerful tool for simulating CME and RDME,
- Demonstration of several models shows convergence with theoritical data for pyLM

Questions ?

open_board